Intelligent Traffic Management Using Fuzzy Logic and Machine Learning

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Project Proposal Report

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DECLARATION

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ABSTRACT

Modern cities continue to struggle with urban traffic jams because vehicle flow remains dynamic and unpredictable. Traditional fixed-time signal systems cause extended delays and safety issues since they are unable to adjust to real-time events like unexpected traffic surges or the arrival of emergency vehicles. To get around these restrictions, this paper proposes an Intelligent Traffic Management System that combines FL and ML. The suggested system uses real-time data processing from computer vision modules, RFID technologies, and infrared sensors to dynamically optimize signal timings. The fundamental novelty is the application of FL to address the basic error of traffic data. Low, medium, and high density are examples of fuzzy linguistic phrases that are created from raw sensor inputs. Lane priority are evaluated by an extensive collection of IF-THEN rules, which produce fuzzy recommendations that are subsequently defuzzed into exact commands for signal duration modifications. The system can learn from past traffic patterns and enhance its decision-making logic over time due to the incorporation of machine learning. The RFID-based EVP protocol, which gives emergency vehicles instant green light early warning represents an important part of the system. Real-time monitoring via a web dashboard is ensured via cloud communication via AWS IOT with MQTT protocol. An MCU with LED signal configurations is used in a physical prototype to evaluate the system's functioning and show real-time operation. According to experimental findings, the priority scoring algorithm minimizes idle time for vacant lanes while efficiently allocating longer green durations to high-density lanes.

Keywords – Fuzzy Logic, Machine Learning, Traffic Management, Adaptive Signal Control, Emergency Vehicle Prioritization.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ML	Machine Learning
FL	Fuzzy Logic
RFID	Radio-Frequency Identification
AI	Artificial Intelligence
LED	Light Emitting Diode.
ITMS	Intelligent Traffic Management System.
IR	Infrared
DFD	Data Flow Diagram
FLC	Fuzzy Logic Controller
EVP	Emergency Vehicle Prioritization
MCU	Microcontroller Unit (ESP32)
IOT	Internet of Things
YOLO	You Only Look Once
DE	Differential Evolution
MQTT	Message Queuing Telemetry Transport
AWS	Amazon Web Services

1. INTRODUCTION

Traditional fixed-time traffic signal systems are severely challenged by the dynamic and unpredictable nature of urban traffic, which frequently leads to high congestion, extended travel times, and increased pollution. Because they follow strict, predetermined schedules, these traditional systems lack the inherent flexibility needed to adapt to real-time events like unexpected emergency vehicle arrivals or abrupt traffic surges. Thus, efficient traffic flow results from the incapacity to manage the complex, multi-variable scenarios of contemporary road networks. While some modern technologies have provided basic adaptive control, many are still limited in scope, concentrating on isolated crossings or depending on crude sensor inputs that are unable to offer a thorough, complete picture of the present traffic conditions. The urgent need for a more advanced, adaptable, and extensible ITMS (Intelligent Traffic Management System) that can adjust to a variety of real-world circumstances is highlighted by this technology gap.

An FLC (Fuzzy Logic Controller) for adaptive signal control is being developed to increase the effectiveness of urban intersections in order to satisfy this requirement. The proposed ITMS dynamically modifies traffic signal timings using real-time data from a network of RFID (Radio-Frequency Identification) and IR (Infrared) obstacle sensors. The use of FL (Fuzzy Logic) to handle the inherent imprecision and uncertainty usually present in raw traffic data is a major improvement of this system. The system uses a set of "IF-THEN" algorithms to assess and determine lane priorities by translating sensor inputs into fuzzy linguistic variables like "low," "medium," and "high" traffic density. In order to ensure a more seamless transition between traffic phases, this logic-driven process produces a fuzzy recommendation that is then de-fuzzified into explicit orders to alter signal durations in real time.

The system's technological architecture includes an MCU (Microcontroller Unit (ESP32)) for data processing and physical LED (Light Emitting Diode) setups for signal simulation. Data collection is primarily handled by IR sensors, which detect the presence and density of vehicles at the stop line, while ML (Machine Learning) algorithms are integrated to allow for continuous self-improvement. This AI integration enables the system to evaluate historical data and gather knowledge from changing traffic patterns over time, resulting in predicted adjustments that further reduce delays. By functioning within an IOT (Internet of Things) framework, the system ensures

that different modules can successfully interact, resulting in a robust platform for context-aware signal optimization that outperforms standalone, non-communicative traffic controllers.

Furthermore, the system features a dedicated EVP(Emergency Vehicle Prioritization) protocol, a critical element often overlooked in existing management frameworks. In order to clear the way, this system uses RFID technology to identify incoming emergency vehicles and immediately override the conventional fuzzy rules. This integrated approach provides a comprehensive solution to the shortcomings of existing traffic control methods and was created especially to manage the special and frequently chaotic traffic circumstances typical of Sri Lankan highways. A physical prototype with an Arduino-based MCU and LED signals will be built in order to offer a tangible proof of concept and show real-time operation. This all-encompassing strategy seeks to provide a traffic management solution for contemporary metropolitan settings that is safer, smarter, and incredibly responsive.

1.1 Background and Literature Survey

Urban traffic congestion constitutes a major worldwide issue that causes longer travel times, more fuel use, and more pollution of the environment. The erratic swings of contemporary metropolitan traffic are not taken into account by conventional traffic signal systems, which operate on set time intervals. This inflexibility frequently causes needless delays and ineffective use of the roads. Research on ITMS focuses on developing adaptable frameworks with real-time responsiveness in order to overcome these constraints. An FLC intended for Adaptive Signal Control represents a key element of this invention. In contrast to static systems, this controller optimizes signal timings using real-time data, specifically addressing the extremely dynamic and non-linear traffic patterns typical of areas such as Sri Lanka.

In order to provide a self-adapting decision-making process within the MCU, the suggested method sets itself apart by combining ML and FL. In order to determine lane priorities using a sophisticated set of "IF-THEN" algorithms, raw sensor data typically obtained by infrared sensors can be transformed into linguistic variables like "low," "medium," or "high" traffic density. These qualitative evaluations are then converted into exact numerical commands to dynamically modify LED signal durations. To guarantee smooth EVP, the system also integrates RFID technology and IOT connectivity. The fuzzy rules' intelligence increases over time by using a continuous learning

process within an AI framework, which enables the system to enhance accuracy and efficiency based on past traffic behavior. An urban transportation network that is more responsive, intelligent, and flexible will be ensured by this hybrid strategy.

- ☐ **Limited Adaptability (Jafari et al., 2022 and 2021):**
  - Jafari et al. (2022) [1] and (2021)[2] used FL and Lyapunov algorithms to control traffic signals.
  - The lack of adaptive learning in these research limits the system's ability to improve based on previous data.
  - Additionally, because their rule bases are static, they can't evolve over time to accommodate continuously shifting traffic situations.
- ☐ **Focus on Isolated Intersections (Mahmood et al., 2019):**
  - Mahmood et al. (2019) [3] demonstrated a two-stage FL system for a single junction.
  - This strategy seems not ideal for managing complex intersections on a broad scale in real-world scenarios, despite its effectiveness in some cases.
  - The proposed technique aims to manage complex traffic patterns and several intersections.
- ☐ **Lack of Learning Mechanism (Salehi et al., 2014):**
  - Salehi et al. (2014) utilized wireless sensor networks to create a multi-agent system based on FL [4].
  - The research lacks a learning method for improving control logic, similar to previous studies in the industry.
  - The recommended function improves FL rules' intelligence through a learning method.
- ☐ **Reliance on Rigid Logic (Liu et al., 2024):**
  - Liu et al. (2024) introduced an adaptive control approach based on logic principles [5].
  - The proposed method is more secure because it uses FL, which is better suited to handle the inherent gaps and imprecision in real-time traffic data.
  - Although being functional, this method has limitations because it relies on pre-established logic rather than a fuzzy system's ability to handle confusing data.
- ☐ **Static Rule Limitations in Multi-Agent Systems (Salehi et al., 2014):**
  - Research in the TLCSBFL plan [6] establishes a foundation for using FL in multi-agent autonomous traffic control using wireless sensor networks.

- The study's reliance on a fixed rule base, which is unable to adapt or learn from past traffic patterns, is one of its main limitations.
- The suggested ITMS enhances this by including ML and adaptive learning to continuously update "IF-THEN" rules, enabling the system to adapt to changing traffic conditions over time.

☐ **Constraints in Emergency Priority and Queue Management (Hanif et al., 2025):**

- An adaptive fuzzy logic system for isolated intersections with a focus on vehicle queue length and waiting time is shown in the paper by Hanif et al. [7].
- Although this study presents EVP logic, it mainly uses queue length settings and MATLAB simulations, which might not adequately represent the intricacy of real-world hardware interactions.
- By developing a physical hardware prototype with RFID technology and infrared sensors, the ITMS improves this idea and offers a real-time, context-aware solution customized for Sri Lankan Road conditions.

The offered research effectively overcomes the limitations noted in existing literature by combining FL's risk-handling capabilities with machine learning's predictive and adaptive characteristics. By moving away from static rule sets and allowing for continuous learning, architecture achieves intelligence that traditional systems do not have. This dynamic approach addresses the drawbacks of traditional traffic management methods. Furthermore, the system is useful since it incorporates RFID-based emergency vehicle prioritizing and is designed specifically for Sri Lanka's unique traffic conditions. These improvements ensure that the proposed approach remains a more reliable and efficient tool for addressing true traffic congestion issues.

1.2 Research Gap

Conventional signal systems, which are frequently reliant on fixed schedules or limited sensor data, have significant difficulty responding to the dynamic and unpredictable nature of urban traffic congestion. This lack of flexibility frequently leads to inefficiencies, increased congestion, and long wait times. Although AI approaches have been integrated into many modern systems to create adaptive controllers, the majority of solutions are designed for specific scenarios and lack the resilience needed to manage the complexity of real-world traffic flows. For example, the approach in [8] optimizes signal timing based on flow rates from simple counters but fails when unusual occurrences occur, such as the need for EVP. Furthermore, studies that focus on a single,

specialized junction type have limited application to complex road networks. Much of the existing literature focuses on particular aspects of traffic management rather than an overall approach. While complicated reinforcement learning models seek to reduce delay at isolated junctions, they frequently necessitate the gathering of complex features, which can result in performance degradation as data volume increases. In contrast to works like [10], these models do not specifically examine how a system can identify lanes with the highest traffic to prioritize them. Similarly, [9] investigates ML-based flow optimization for a single, one-line junction. While effective for that specific arrangement, a more robust system is necessary to handle a variety of junction types, including three- and four-way intersections, indicating a substantial gap in the field.

Technically, a solid system utilizing FL and ML to manage traffic in real time while addressing context-aware scenarios is absent. There has been little investigation into employing a combined technique to manage multiple types of junctions at the same time, which could offer more accurate data for management over a larger road network. Most FLC implementations lack a defined technique for instantaneous EVP, which raises safety and efficiency concerns. Furthermore, the lack of a clear strategy for analyzing and prioritizing certain high-traffic lanes, rather than simply minimizing overall delay, remains a limiting factor in enhancing responsiveness amid high traffic.

The industry also faces substantial problems in terms of scalability and hardware integration. For example, the agent-based fuzzy system described in [11] identifies a "rule-base explosion" problem, in which the number of FL rules rises exponentially with the number of crossings, resulting in enormous computing costs. While that study implies future ML integration to predict traffic flow, the ITMS fills this gap by using adaptive ML to constantly refine FLC knowledge. This prevents computational bloat and keeps the system scalable. Furthermore, unlike [11], which relies on expensive communication networks, this version uses low-cost components such as the MCU and IR sensors, giving a low-cost, high-performance option ideal for Sri Lankan Road conditions.

Another important gap occurs in specialized vehicle handling and sensor accuracy. The optimization approach mentioned in [12] focuses solely on lowering average vehicle delay and does not include an EVP mechanism. In contrast, this system employs robust hardware-software fusion via RFID tags to provide emergency vehicles with immediate preemption. While [12] employs a "stepwise" optimization approach to identify fixed parameters for various traffic

circumstances, the ITMS provides continuous adaptability via real-time multi-sensor data integration. This study expands on the simple vehicle counts used in typical simulations by using inputs from IR sensors for vehicle density and RFID for identity-based priority to build a context-aware traffic management strategy.

The work aims to bridge these gaps by developing a fully functional ITMS capable of dynamically adjusting signal timings. Stability can be enhanced by integrating numerous components, such as parking and violation detection data. Combining ML with FL results in a flexible control system suited to handle real-world scenarios, such as EVP detection via RFID [10]. Moving beyond the single-junction focus of prior efforts [9], this architecture is meant to support a variety of junction types via an MCU-based IOT framework. Unlike techniques based on reinforcement learning, this strategy expressly employs FL to prioritize high-traffic lanes, guaranteeing that the busiest parts receive the necessary attention via LED indicators and sensors such as infrared [10]. Addressing these weaknesses improves traffic flow and safety results.

1.3 Research Problem

Urban traffic congestion remains a major global issue, contributing significantly to environmental pollution, wasteful fuel usage, and considerable economic deficits. The major challenge is that real-time traffic flows are inherently dynamic and unpredictable. Conventional traffic signal systems, which often use static timings or pre-programmed schedules, lack the ability to respond to these variations. Wait times are too long because these legacy systems lack the ability to respond to unanticipated factors such as traffic accidents, unexpected vehicle surges, or the urgent arrival of emergency services. This inherent inflexibility exacerbates traffic, particularly during peak hours when rapid response needs to happen to preserve order on the metropolitan road network.

While several modern automated techniques have been developed, many still rely on crude sensor inputs that are unable to handle complicated, multivariable situations. The propensity to handle intersections separately, frequently ignoring the detrimental impacts on the larger urban network, is a major drawback of current systems. Furthermore, the practical implementation of many existing algorithms in complicated metropolitan contexts is hampered by their lack of the computing efficiency needed for large-scale, real-time applications. An ITMS that can smoothly combine a variety of real-time data inputs from different sources such as vehicle density and

emergency vehicle detection into accurate and efficient signal control actions is desperately needed.

The creation of an FLC for adaptive signal control remains the main goal of this study in order to address these limitations. By transforming raw data from infrared sensors into fuzzy language concepts that appropriately characterize different traffic conditions, the system aims to manage the ambiguity and uncertainty of traffic patterns. The controller assesses lane priorities and produces a fuzzy recommendation for ideal signal intervals by using a complex set of "IF-THEN" logic rules. Green light lengths can be dynamically adjusted according to this procedure, which guarantees that the junction runs at maximum efficiency based on real demand rather than theoretical schedules.

The integration of ML into the real-time FL framework to enable ongoing self-improvement and adaptation marks a fundamental innovation of the suggested design. By the use of machine learning, the system may gradually improve its decision-making logic, guaranteeing that the FLC continues to function well even when traffic patterns change. In order to guarantee strong EVP, the system also uses RFID technology, which enables instantaneous signal preemption in the event that a priority vehicle is identified. This functionality is specifically designed to accommodate the unique and often chaotic road conditions found in Sri Lanka, where standard lane discipline and predictable vehicle behavior are frequently absent.

In order to show that the IOT framework appears practically feasible, the last phase of this research entails building a physical prototype using a high-performance MCU. This hardware model replicates and test the dynamic signal changes in real-time using LED indicators and infrared sensors for detection. The study shows how an accurate and automated ITMS can accomplish smarter traffic management by offering tangible proof of concept. Ultimately, this comprehensive approach seeks to reduce congestion, minimize transit delays, and significantly enhance overall road safety through the synergy of AI and embedded hardware.

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this research aims to optimize urban traffic flow by designing, developing, and implementing an intelligent and adaptive ITMS that makes use of the complementary skills of ML and FL. In order to adapt signal timings to Sri Lanka's particular road circumstances, the system processes multi-modal real-time data, such as vehicle density and the presence of emergency vehicles. The project's ultimate goals are to reduce traffic, improve road safety, and provide effective EVP within updated urban structures.

1.4.2 Sub Objective

Sub-objectives that have been identified to achieve the main objective are:

- **To create a reliable system for gathering and combining real-time traffic data.**

Develop a comprehensive hardware-software system that uses RFID and infrared sensors to continuously collect data on lane occupancy and vehicle counts. Preprocessing the raw input to guarantee data integrity and suitability for high-level analysis by the FLC and ML modules makes up of this goal.

- **To Design and Implement an Adaptive FLC for Dynamic Signal Optimization**

Construct the core logic unit that can translate numerical sensor data into fuzzy language phrases like "Low," "Medium," and "High" traffic density. This entails developing a specific "IF-THEN" rule foundation to assess lane priorities and produce accurate suggestions for increasing or decreasing green light durations.

- **To Integrate ML Algorithms for Predictive Traffic Behavior Analysis**

Enhance the system's intelligence by teaching supervised learning models to recognize traffic flow patterns and forecast possible areas of congestion. By improving fuzzy rules based on expected vehicle surges, this predictive layer enables the ITMS to move from reactive control to proactive management.

- **To Develop a Multi-Modal Protocol for Emergency Vehicle Prioritization**

Develop a high-priority subsystem that recognizes approaching emergency vehicles using RFID tags (13.56 MHz) and sensor fusion. In order to clear a safe path and shorten essential response times, the goal is to start rapid signal preemption through the FLC.

- **To Build a Functional Hardware Prototype and IOT Monitoring Interface**

Build a physical proof-of-concept by simulating real-world junction logic and controlling LED signal indications with an MCU. For real-time monitoring and remote command override, this involves setting up an IOT gateway to publish traffic data to a cloud-based monitoring platform every 5 seconds.

- **To Validate System Performance through Simulation and Comparative Metrics**

Utilizing a simulated environment based on Sri Lankan road networks, assess the integrated ITMS's efficacy. Key performance indicators, such as reduced average vehicle wait times, higher traffic throughput, and the EVP protocol's success rate, will be utilized to evaluate performance.

2. METHODOLOGY

2.1 System Overview and Methodology

The structured, multi-layered architecture of the suggested ITMS provides to convert processing environmental data into intelligent traffic control measures. The process starts with the Sensors & Devices layer, where data is continuously collected from the physical world, as shown in the high-level system overview. While RFID readers scan the distinct tags attached to emergency vehicles, infrared sensors are positioned strategically to identify the presence and density of vehicles. To guarantee context-aware decision-making, the system also incorporates data from the parking and violation modules. The Fuzzification & Preprocessing stage receives this raw data and transforms numerical quantities, like vehicle counts, into FL language phrases like "Low," "Medium," or "High" flow. This stage needs to be maintained for controlling the inherent unpredictability of real-world traffic patterns, especially in Sri Lanka's often unsafe road conditions.

The Fuzzy Inference Engine and Rule Base, the central intelligence hub, collects the preprocessed data. Here, the algorithm evaluates lane priorities using a wide range of "IF-THEN" criteria. For instance, a lane with "Low" density (a score of 40) will be given less priority than a lane with

"High" density (a priority value of 100). The EVP logic quickly interrupts regular cycles to give the emergency lane a green light if an emergency vehicle is identified using RFID. After this assessment, the Defuzzification & Command Generation module converts these hazy suggestions back into exact numerical commands, such 5, 8, or 12-second green light durations.In the end, the MCU carries out these commands, updating the digital dashboard and managing the hardware prototype's physical LED traffic lights. (*Figure 2.1)*

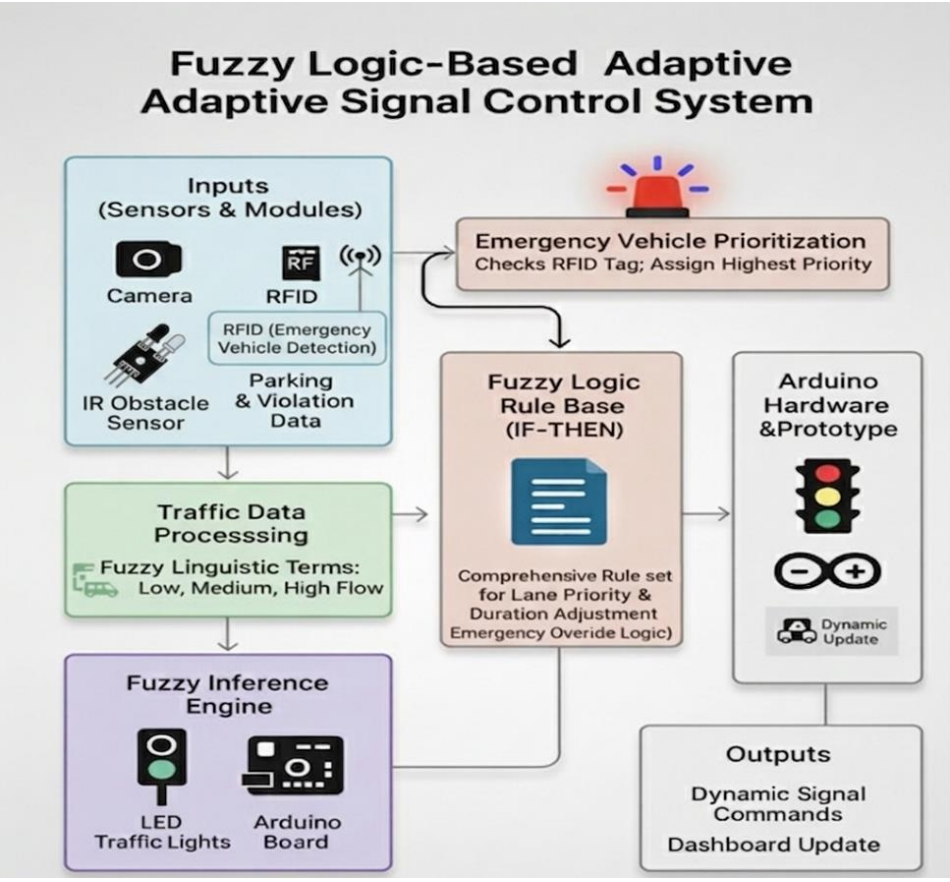

Figure 2.1 Overall Component System Diagram

The complete DFD provides the finest understanding of the internal flow of information and the system's capacity for self-improvement. This framework's continuous and circular flow guarantees that the system will continue to react to changes in real time. Every five seconds, the data acquisition layer uses IOT to publish identification and traffic density data to the cloud. This enables remote command overrides for testing and continuous AWS (Amazon Web Services) IoT monitoring. The Adaptive Learning Module, which use machine learning to examine past and

present feedback from intersection outcomes, is a crucial part of this system. This guarantees that the FLC is dynamic and gradually improves in efficiency.

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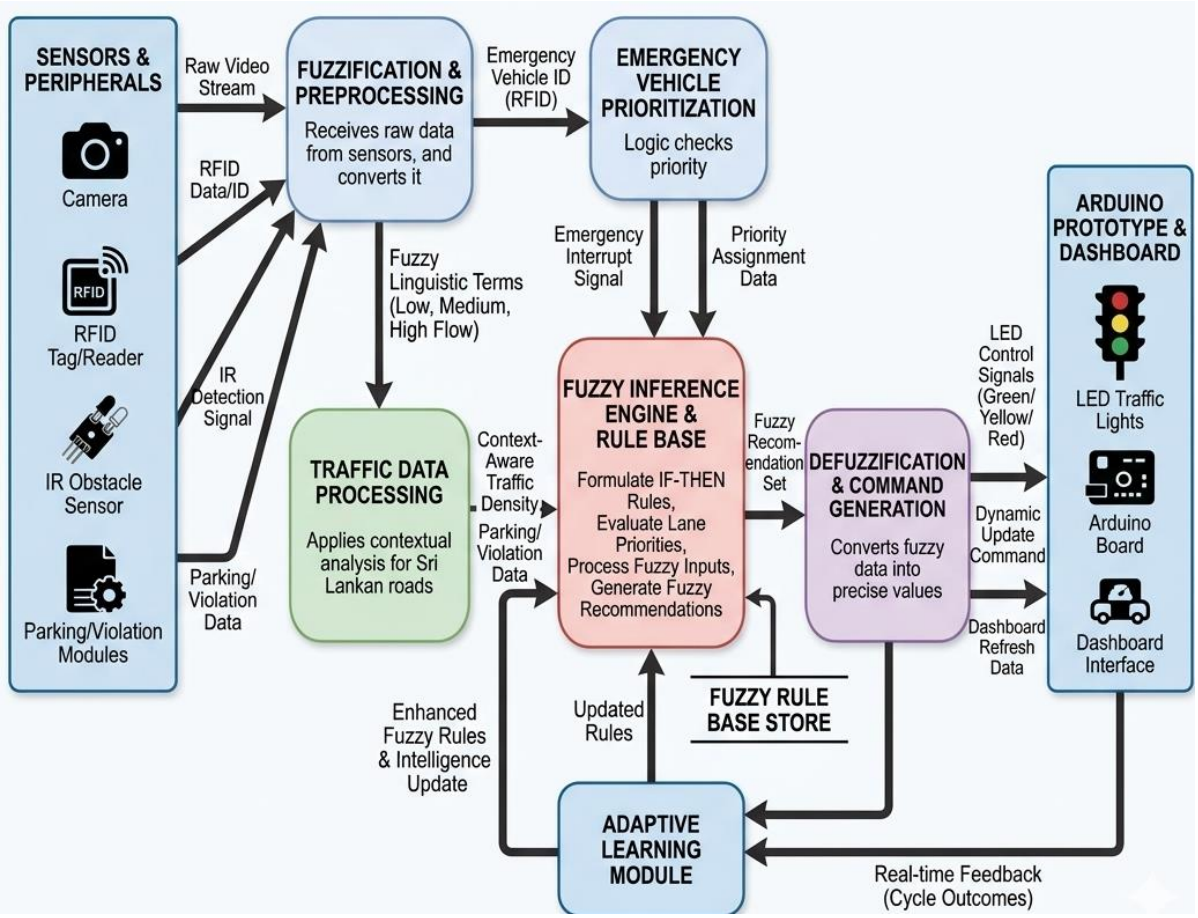

Figure 2.2 Data Flow Diagram (DFD)

2.2 System Requirements

2.2.1 Functional Requirements

- **Continuous Real-Time Data Acquisition**
  The system must collect traffic information from all integrated sensors on an ongoing basis in order to offer a continuous stream of data for decision-making.
- **Conversion of Raw Sensor Data into Fuzzy Terms**
  It must convert unprocessed numerical sensor data into descriptive fuzzy categories to enable to simplify traffic condition analysis and processing.
- **Execution of Logical "IF-THEN" Rule Base**
  To ensure to correctly determine the priority levels for various traffic lanes, the controller must apply particular "IF-THEN" logic rules.
- **Generation of Precise Signal Control Commands**
  The system must generate accurate results instructions to dynamically modify the timing of traffic signals, ensuring the most efficient flow of vehicles.
- **Integration of Adaptive Learning Capabilities**
  It has to contain learning algorithms that enable the system to continuously improve its intelligence and increase the accuracy of its fuzzy rules over time.
- **Identification and Prioritization of Emergency Vehicles**
  To enable quick and secure transportation through the junction, the system must be able to identify emergency vehicles and promptly give them signal priority.
- **Real-Time Demonstration via Hardware Prototype**
  A physical model must be used to provide a clear representation of responsive signal changes, showing the system's ability to function in practical situations.

2.2.2 Non-Functional Requirements

- **Reliability**

Signals must be regularly updated by the traffic control system in a precise and safe way. The logic utilized for data analysis and signal modifications needs to be extremely reliable because mistakes could cause serious traffic jams or accidents. In order to assure stable operation at all times, the FL rules are rigorously tested in a variety of contexts.

- **Security**

The importance of traffic control makes it imperative to safeguard the system from unauthorized access. To stop malevolent manipulation, every communication channel from the sensors to the main controller must be protected. Strong authentication and access control systems guarantee that the system can only be monitored or manually modified by authorized personnel.

- **Scalability**

To handle a vast network of traffic signals, the architecture needs to be able to go beyond a single intersection. Without sacrificing processing speed, the system is built to manage an increasing number of sensors and data inputs. As the urban network expands, this flexibility makes it simple to incorporate new geographic areas.

- **Performance**

Effective real-time management requires high-speed processing. The system must to gather information, use FL logic, and send out signal commands almost immediately. Quick response times guarantee that signal updates occur fast enough to respond to abrupt changes in traffic conditions and avoid delay.

- **Availability**

The system must run constantly in order to guarantee a continual flow of traffic. Important components have built-in redundancy to avoid significant delays brought on by possible breakdowns. A fallback mode maintains fundamental traffic control by automatically switching to a fixed timer plan in the case of a system fault.

- **Usability**

Traffic police officers and maintenance personnel must have a simple user interface to allow them to view data and take appropriate action. To enable quick analysis and error-free operation, the dashboard needs to be clearly developed. This emphasis on user friendly design lowers the requirement for intensive training and boosts productivity.

2.2.3 Tools and Technologies

The research details the use of several tools and technologies, each of which makes a unique contribution to the development and functioning of the system.

Backend & Intelligence

- **Python 3.11 / 3.12:** It serves as the key language for backend development and fuzzy logic processing. It performs system integration and complex data calculations.
- **Fuzzy Logic Controller (FLC):** Utilize language characteristics such as "Low" or "High" to assess lane priorities. It employs "IF-THEN" principles to convert sensor data into signal decisions.

ML & Computer Vision Libraries:

- **YOLOv8 & PyTorch:** Powered the real-time vehicle detection and counting system. These frameworks enable fast object detection and model training.
- **OpenCV:** Controls image processing activities and real-time camera feed inputs. It refers to essential for pre-processing visual data before detection.

Data Sources

- **Sri Lankan Traffic Dataset:** Trains using local traffic film with manual annotations. This ensures that the model correctly classifies regional vehicle categories such as local buses, taxis, and ambulances.

Web Dashboard (Frontend)

- **Flutter SDK 3.16+ and Dart 3.2+:** Used to create a dynamic real-time visualization dashboard. It displays real lane priority and signal durations on multiple web platforms.
- **Google Chrome:** Provides as the primary runtime environment for designing and testing web dashboards. It provides excellent performance for real-time data display.

Cloud & Data Management

- **AWS IoT Core & MQTT:** It enables secure, real-time messaging between devices and the cloud. The MQTT protocol provides low latency data posting every 5 seconds.
- **AWS DynamoDB:** A NoSQL database for storing sensor inputs and historical signal data. It offers excellent scalability for logging real-time traffic events.

Hardware & Embedded Systems

- **Arduino / NodeMCU (ESP8266):** The central microcontroller serves for hardware prototyping. It manages sensor inputs and implements physical traffic light control logic.
- **IR Obstacle & RFID (13.56 MHz):** IR sensors detect vehicle density, whereas RFID tags identify emergency vehicles. These elements cause immediate preemption for high-priority lanes.
- **LEDs & Serial Communication:** LEDs simulate physical signal changes during the prototype demonstration. Serial commands enable manual override and system debugging during testing.

Version Control

- **Git / GitHub:** Used for source code management and collaborative development. It keeps track of all changes and provides a history of the project's evolution.

2.3 Commercialization aspects of the product

The final product appears as a high-value, low-cost urban traffic management solution that is especially suited for the Sri Lankan market in order to solve the commercialization aspects of the ITMS. The main selling point is the use of standard parts like MCU and IR sensors in an affordable architecture to replace costly, imported traffic systems. Focusing on a "hardware-software fusion" that combines ML and FL, the system provides a scalable substitute for conventional infrastructure. Government stakeholders are the first commercial target, especially law enforcement and local governments. Police personnel are able to remotely monitor traffic congestion and prioritize lanes or emergency vehicles since the FLC outputs are linked into a digital dashboard. Manually traffic direction, which often proves ineffective and prone to human mistake during peak hours, represents no longer necessary as a result.

A hardware prototype's cost-effectiveness is a crucial component of commercialization. ITMS uses a simplified list of components that keep production costs very low, whereas global smart traffic solutions can cost millions of dollars (Table 2.1). These numbers reflect the initial prototype expenses, but it's crucial to remember that when mass production progresses, economies of scale and bulk purchasing will probably result in even lower unit prices. Even while the overall cost would go up if high-performance parts like cameras for sophisticated visual identification were added, the system as a whole is still far less expensive than standard infrastructure.

Item	Quantity	Unit Price (LKR)	Total (LKR)
PCF8574	2	Rs. 900.00	Rs. 1,800.00
SN74HC595	2	Rs. 120.00	Rs. 240.00
MFRC522 (RFID Module)	4	Rs. 260.00	Rs. 1,040.00
IR Sensors (Infrared Obstacle Avoidance)	12	Rs. 167.00	Rs. 2,004.00
ULN2003A (SOP-16 Package)	2	Rs. 120.00	Rs. 240.00
LEDs (Red/Yellow/Green - Assorted)	12	Rs. 3.00	Rs. 36.00
Resistors (For LEDs)	12	Rs. 1.00	Rs. 12.00
Breadboard	2	Rs. 130.00	Rs. 260.00
Jumper Wires (100pcs pack)	3	Rs. 240.00	Rs. 720.00

Table 2.1: Prototype Component Cost Breakdown

The IOT-based prototype shows how the system works in a connected urban setting by gathering real-time environmental data and sending it to a cloud-based dashboard. This makes data-driven decision-making and ongoing evaluation possible. The logic can be modified to include high-sensitivity capture cameras if this system is designed for practical use. The FLC might use the high-resolution visual data from these cameras to better analyze traffic patterns and improve the "IF-THEN" algorithms for even more accurate signal timing. As infrastructure advances, this shift from simple sensors to complex visual inputs maintains the system's continued value.

The current business model includes a high-value software-as-a-service (SaaS) component in which police officers and urban planners use the IOT dashboard to analyze real-time traffic trends and infraction data published by AWS IoT every five seconds. While the primary value proposition is based on these well-established monitoring and smart city project tools, the architecture is

designed for modular expansion, allowing for the future integration of specialized EVP protocol access and RFID tag services for emergency response agencies as scalable service enhancements.

Scalability remains another commercial advantage, as the system aims to manage a growing number of sensors without reducing performance. The modular architecture allows a city to begin with an individual junction and eventually grow to a complete network. Furthermore, because the ML and FL regulations are tailored to Sri Lankan Road circumstances, there is a great chance to export this technology to other South Asian countries with similarly chaotic traffic scenarios. This regional adaptability makes the product an appealing investment for technology companies trying to address global South urbanization issues. Finally, the ITMS's competitive advantage is its capacity to reduce economic losses resulting from fuel waste and operational delays. By increasing traffic speed and decreasing vehicle idle time at intersections, the system provides a clear return on investment for the national economy.

2.4 Testing & Implementation

2.4.1 Implementation

The ITMS can be executed using a multi-layered strategy that combines advanced cloud and intelligence modules with physical hardware. An ESP8266 NodeMCU serves as the key microcontroller unit in the hardware prototype's basic construction (Figure 2.4.1). This device connects with MFRC522 RFID modules to identify emergency vehicle tags and a network of twelve infrared sensors to track vehicle density. PCF8574 I/O expanders and SN74HC595 shift registers are used to manage the large number of inputs and outputs needed for a sophisticated junction simulation. Twelve LEDs that show the red, yellow, and green signal states for several lanes may be controlled by the MCU as a result to these parts. The hardware logic may be evaluated against concurrent traffic demands from many directions thanks to the physical layout's simulation of a four-way junction.

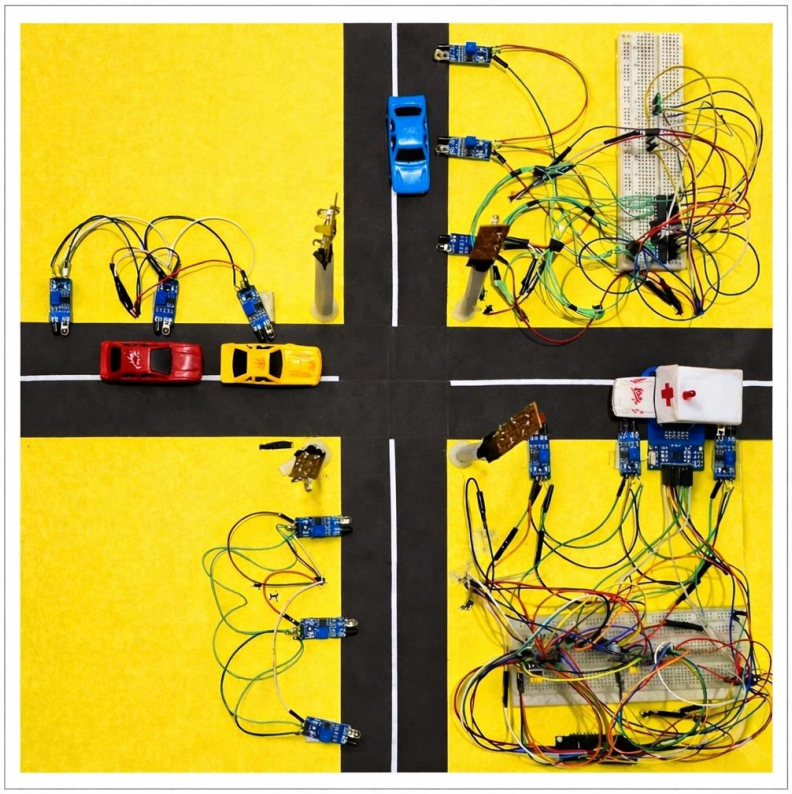

Figure 2.4.1 Fuzzy Logic Controller for Adaptive Signal Control IoT Hardware Prototype

Python 3.12, which serves as the link between raw data and decision-making logic, is usually used in the software implementation. The Fuzzy Logic Controller is implemented by the system using the scikit-fuzzy package. This entails mapping clear numerical data from infrared sensors to verbal phrases like "empty," "low," "medium," and "high." YOLOv8 and OpenCV-powered computer vision modules simultaneously process visual feeds to confirm car counts, guaranteeing great accuracy even in Sri Lanka's typical chaotic road circumstances. An adaptive learning module that evaluates real-time feedback to improve the IF-THEN rules stored in the system's rule base enhances the core intelligence. This provides a hybrid intelligence layer in which the fuzzy logic engine determines the final signal timing after computer vision verifies the physical sensor data.

The web-based visualization dashboard, created with the Flutter SDK and Dart, represents an important part of the implementation (Figure 2.4.2). This dashboard offers a real-time interface for law enforcement and traffic officials to monitor the entire system. The interface, as shown in the IoT Traffic Control dashboard, displays the active lane status, current vehicle counts for North, South, East, and West directions, as well as the traffic signal's current state. The MQTT protocol-

based interface with AWS IoT Core ensures that data is synchronized every five seconds between the physical prototype and the web dashboard. This configuration enables remote monitoring and manual serial command overrides, demonstrating a complete proof of concept for a linked urban traffic environment.

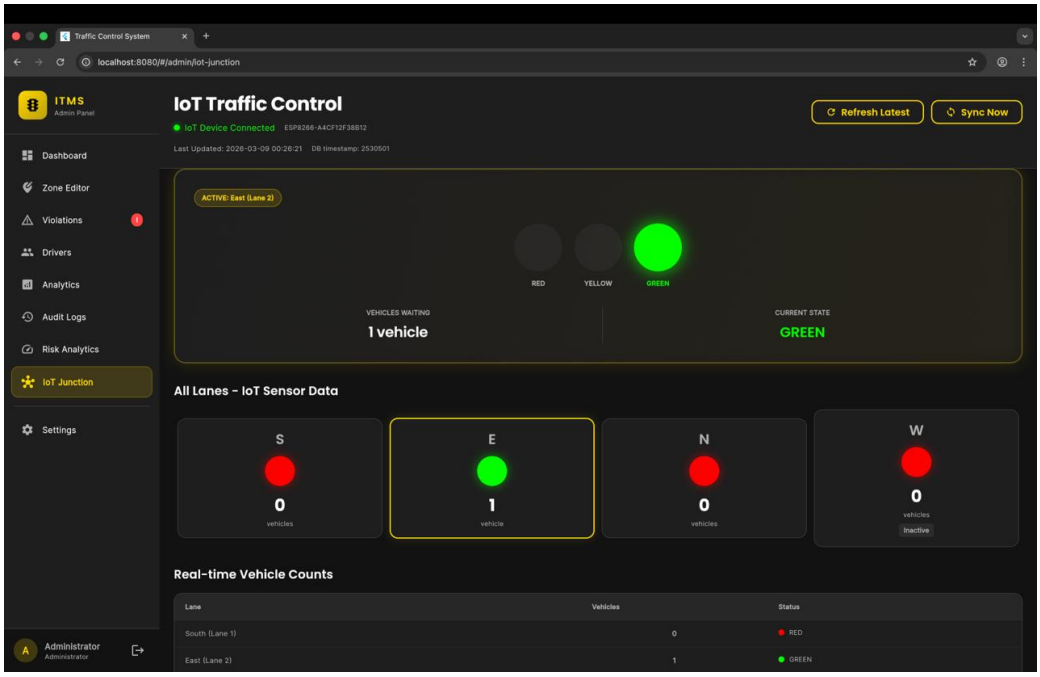

Figure 2.4.2 Fuzzy Logic Controller for Adaptive Signal Control Dashboard

The system architecture's structural design relies on a decentralized processing approach, with heavy computing duties, such as vehicle detection and fuzzy inference, handled by a local workstation or cloud server, and the MCU focusing on real-time hardware execution. Data can spread from the IR sensors and RFID modules to the NodeMCU, which converts the signals into JSON format. This information gets released to a specified MQTT topic on AWS IoT Core. A Python script acting as a subscriber collects this data, processes it using the YOLOv8 detection model and the fuzzy logic engine, and then sends the optimum green light duration back to the device. This continuous connectivity ensures that the physical signals are always powered by the most recent intelligence extracted from the multi-module input system.

The priority score system serves as the central logic for lane selection in the FLC. Each lane receives a priority score depending on vehicle counts, which ranges from 10 for an empty lane to 100 for a high-density lane. This scoring technique are essential for handling complicated junctions

where many lanes may be experiencing varied amounts of congestion simultaneously. The computer uses these scores to determine which lane should get the green light first and for how long. For example, a lane with a "High" density (score of 100) is automatically assigned a 12 second green duration, whereas a lane with a "Low" density (score of 40) is only given 5 seconds. This dynamic adjustment saves time on empty roads and significantly decreases traffic jams.

Emergency Vehicle Preemption (EVP) are constructed into the implementation as a priority signal in the software code. The MFRC522 RFID reader begins up to scan for certain approved tags assigned to ambulances. When a tag has been determined, the NodeMCU transmits an immediate priority signal to the Python controller. The fuzzy logic engine then overrides the current signal cycle, causing an immediate switch to green for the lane where the emergency vehicle can be identified. This override is logged in AWS DynamoDB for auditing purposes and immediately reflected on the Flutter dashboard. This specialized implementation ensures that life-saving vehicles are not delayed by ordinary traffic control logic, providing a reliable solution for emergency response in cities.

The backend data management layer uses AWS DynamoDB to store historical traffic statistics and system logs. Every state change, vehicle count, and emergency override is marked with a timestamp. The scikit-learn adaptive learning module then analyzes this historical data to identify trends. By reviewing these records, the system may detect peak traffic hours and repeated traffic points, allowing it to modify the fuzzy logic engine's membership functions. This use of a feedback loop implies that the system improves over time, responding to the accurate cycles of the junction it manages.

The integration of the computer vision pipeline into the entire design serves as a failsafe for physical sensors. OpenCV can be utilized to record individual frame images from junction cameras, which are next processed by the YOLOv8 model trained on Sri Lankan vehicle datasets. The use of this multi-source validation logic ensures that signal control commands delivered to hardware are always based on correct and confirmed data.

The implementation of the MQTT protocol using AWS IoT Core ensures communication security and low latency performance. The system authenticates the NodeMCU using X.509 certificates, ensuring that only authorized hardware can publish traffic data or receive control commands. By creating a continual keep-alive connection, the system ensures that the 5-second update interval is

carefully followed, which is crucial for synchronizing the real-time dashboard and the actual traffic light demonstration.

The final assembly and integration testing of the prototype include synchronizing the serial communication between the Python backend and the hardware LED display. Serial commands relate to specific signals. The implementation includes a manual override capability in the Flutter dashboard, which allows a police officer to take control via the AWS cloud if an unexpected issue develops on the road. This combination of autonomous fuzzy-controlled logic and manual human intervention creates a comprehensive management framework that balances AI-driven efficiency with human oversight, achieving the main goals of Intelligent Traffic Management System research.

2.4.2 Testing

The testing phase assesses the integrated ITMS's functional dependability and performance efficiency. The testing approach has been split into three stages: unit testing of individual components, integration testing of the hardware-software interface, and simulation-based performance analysis. The initial unit tests focus on calibrating IR sensors to ensure accurate vehicle identification and validating RFID tag reading range for emergency vehicle preemption. Serial communication tests confirm that the NodeMCU receives and executes commands from the Python-based FLC without any latency concerns. A thorough inspection of the hardware logic assures that each component of the four-way junction prototype retains electrical stability and signal integrity. Modular validation ensures that each specific component of the circuitry, including the shift registers and I/O expanders, fully functioning before moving on to the computationally complicated testing steps.

2.4.3 Functional Testing

Functional testing focuses on ensuring that each unique capability of the ITMS meets the given requirements. A major source of worry involves the constant real-time data capture and subsequent conversion of raw sensor data to fuzzy words. Tests guarantee that numerical inputs from IR sensors accurately transfer to descriptive categories like "Low," "Medium," and "High." The logical "IF-THEN" rule base is validated by inputting various traffic density scenarios and ensuring that the FLC provides the expected priority scores and signal control commands. This ensures that the dynamic change of traffic signal timings (5s, 8s, or 12s) occurs perfectly in line with the real-time conditions captured at the simulated junction. Every rule in the rule base gets

tested in controlled situations to ensure that the fuzzy inference engine prioritizes lanes with the highest vehicle count while properly scaling green intervals to avoid excessive idling.

Identification and priority of emergency vehicles represents a fundamental functional requirement that has been completely validated. The MFRC522 RFID modules and authorized tags are used during testing to simulate the arrival of an ambulance. The system can be tracked to ensure that detection results in an immediate interrupt signal, ignoring the typical fuzzy logic cycle and providing a quick green light. The integration of adaptive learning capabilities is also validated by feeding historical data patterns into the scikit-learn module to ensure that fuzzy rules change over time. These functional tests ensure that the hardware prototype effectively simulates responsive signal changes in alignment with real-world traffic management requirements.The seamless transition between standard autonomous mode and emergency override mode remains manually monitored to ensure that no logic loops or deadlocks occur during the preemption period, allowing life-saving vehicles to pass through the intersection at any time.

2.4.4 Non-Functional Testing

Non-functional testing assesses the ITMS's operational characteristics, such as dependability, security, scalability, and performance. Reliability testing entails running the hardware prototype for an extended period of time to check that the ESP8266 NodeMCU and the linked sensors continue to function properly without data loss or logic errors. Performance testing analyzes the system's delay, specifically the time it takes to acquire sensor input, process it through the Python-based intelligence layer, and provide a command to the LEDs. To achieve the high-speed processing requirements, the system must maintain a low-latency loop that ensures signal updates respond quickly to sudden shifts in traffic density.

Security and availability are also important aspects of the non-functional assessment. Authentication techniques and MQTT secure communication channels are evaluated to avoid unauthorized access or malicious tampering of traffic signals. Scalability testing assesses the AWS IoT Core and DynamoDB architecture's ability to handle growing volumes of data input from many simulated sensors while maintaining processing speed. In addition, availability gets examined by simulating component failures to confirm that the fallback mode effectively transitions to a fixed-timer plan. Usability testing is undertaken on the Flutter-based dashboard to ensure that traffic police can easily and clearly read real-time data and apply manual overrides.

Stress-testing the cloud connection and database queries confirms the system's capacity to remain available and responsive even during peak data transmission periods, setting a high bar for modern smart city applications.

2.4.5 Integration and System Testing

Integration testing ensures that data flows seamlessly from physical sensors to the cloud-based dashboard. This includes testing the MQTT connection's stability and the accuracy of the data saved in AWS DynamoDB. Specific test cases imitate priority situations, such as the arrival of an ambulance. During these tests, the system ensures that the RFID module sends an immediate interruption signal. This step prompts the FLC to override the current cycle and provide an instant green light to the emergency lane. The dashboard's responsiveness also indicates that signal changes on the hardware prototype are reflected on the web interface in real time. The emphasis here is on the seamless interaction between the local hardware unit, the intermediate cloud bridge, and the final visualization layer, which ensures that no data packets are missed and that the junction state is properly synced across all platforms.

The final level of testing includes a quantitative assessment of the system's effectiveness using simulated traffic conditions. Performance can be calculated using important criteria such as average vehicle wait time, traffic flow, and emergency response time. The priority score system, which assigns values ranging from 10 to 100 according on density, is carefully evaluated to verify that lanes with more congestion continue to get longer green light durations. A relative examination compares the performance of adaptive ML enhanced fuzzy rules to standard static timing schedules. By modeling various junction structures, including three- and four-way intersections, the testing process shows that the technology efficiently reduces delays and enhances overall road safety. These accurate system tests confirm that the combination of machine learning and fuzzy logic offers a clearly better performance. These accurate system evaluations demonstrate that the combination of machine learning and fuzzy logic improves traditional traffic infrastructure in terms of decreasing fuel waste and economic losses.

2.4.6 Environment and Tool Validation

The testing environment makes use of the entire backend and intelligence tool set to check the system's core integration. Python 3.12 can be utilized to execute automated test scripts that check the mathematical outputs of the scikit-fuzzy inference engine. To confirm vehicle detection accuracy and precision, the YOLOv8 and PyTorch models are tested against a locally curated Sri

Lankan traffic dataset. OpenCV can be utilized to ensure that real-time camera feed inputs are properly pre-processed before they enter the detection phase. These technologies enable for a thorough examination of how the system detects and responds to various vehicle kinds, such as local buses, taxis, and ambulances. Every line of code in the intelligence layer are examined to ensure it appropriately manages the asynchronous nature of the sensor inputs and maintains a detailed log of all decision-making actions.

The cloud and data management layers are stress tested to ensure that the AWS infrastructure can handle continuous data posting every five seconds. The MQTT protocol can be verified to assure low-latency messaging between the NodeMCU and the cloud, and DynamoDB is tested for accuracy in logging historical traffic information. The Flutter SDK and Dart environment are used to test the web dashboard's responsiveness across several browser systems, particularly Google Chrome. Finally, version control using Git and GitHub serves to log all testing changes and keep track of the system's evolution during the validation phase. The extensive validation of these digital and physical instruments guarantees that every component of the ITMS functions properly, providing direct validation of the system's practical viability for real-world urban deployment.

2.4.7 Work Breakdown Structure

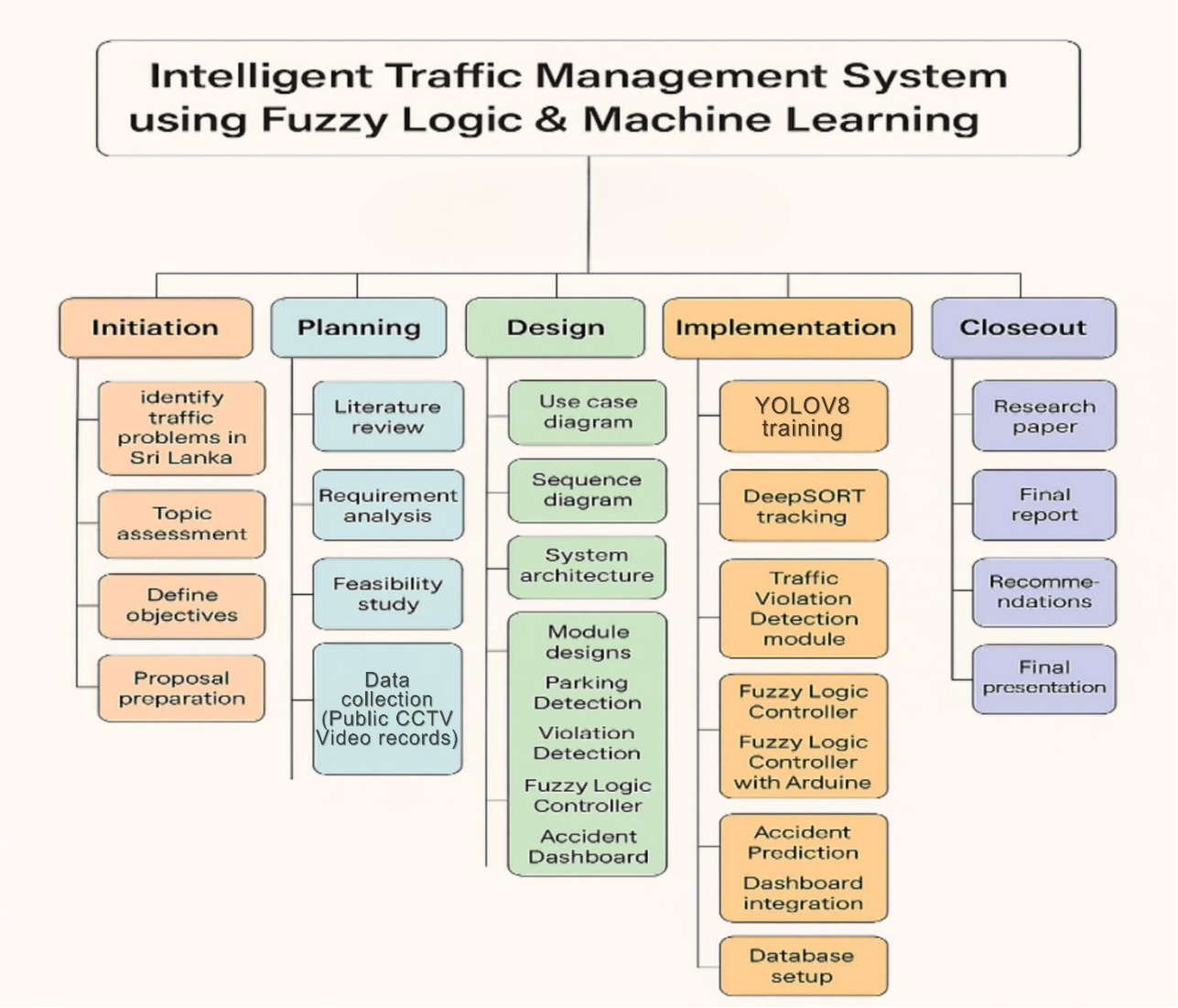

Figure 2.4.3 Work Breakdown.

2.4.8 Gantt Chart

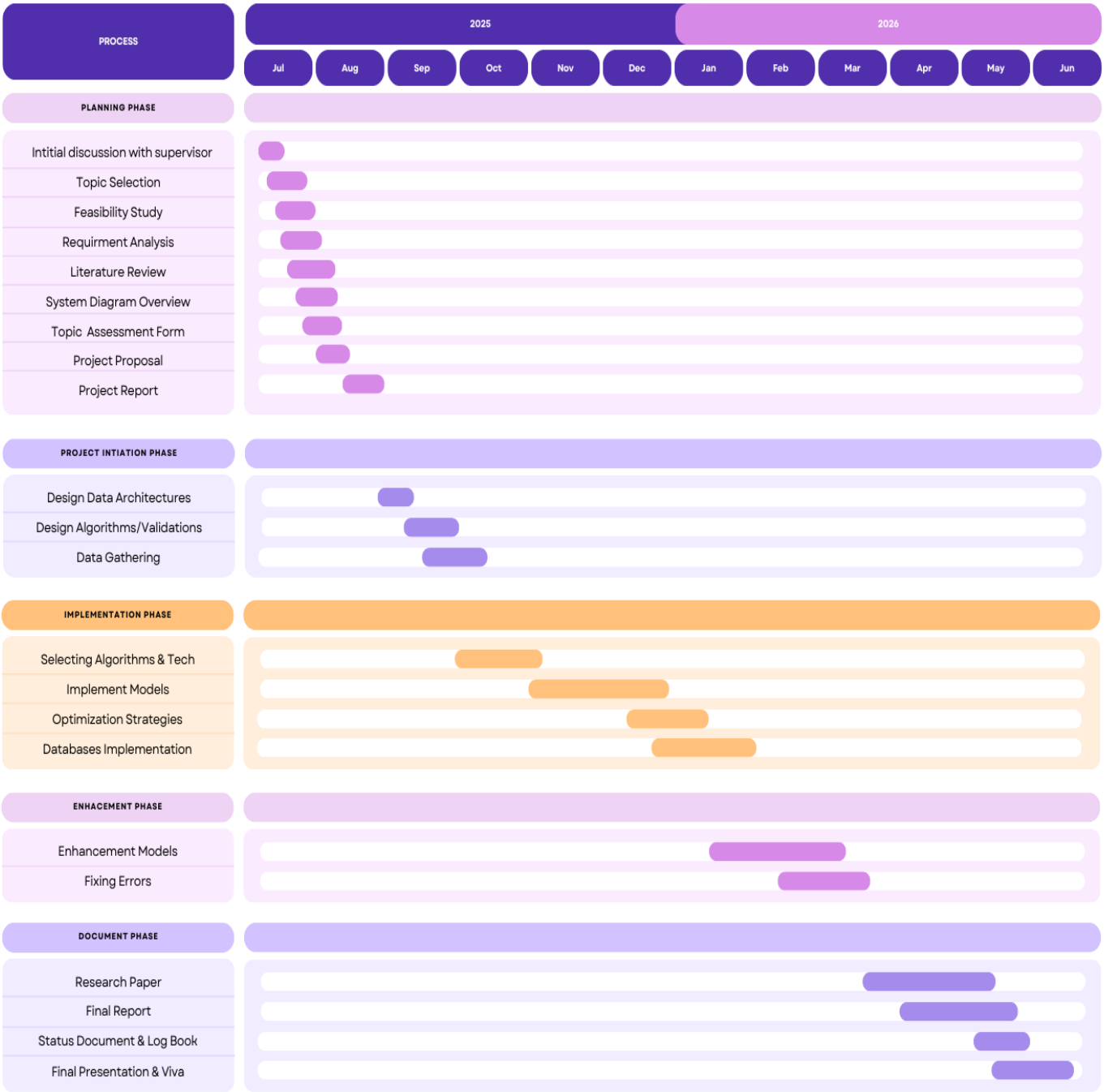

Figure 2.4.4 Gantt Chart

3. RESULTS & DISCUSSION

3.1 Results

The ITMS study provides valuable insights into the effectiveness of merging FL and ML for urban traffic control. When compared to traditional fixed-timer methods, system performance improves significantly in terms of junction flow management. Data from the real hardware prototype demonstrates that the priority scoring algorithm efficiently converts vehicle density into optimal green light intervals. During peak traffic simulations, the FLC correctly identifies high density lanes and increases signal durations to alleviate congestion while maintaining junction equilibrium. The real-time synchronization between the MCU and the AWS dashboard provides traffic administrators with an accurate, low latency picture of intersection dynamics, with updates every five seconds over the MQTT protocol.

Emergency vehicle preemption provides a key success criterion for designed architecture. Testing with RFID tagged automobiles demonstrates that the system regularly activates the EVP protocol with almost immediate response times. When a high priority tag appears the FLC interrupts standard logic and provides an immediate green light, assuring a safe and clear path. This capability addresses the important reaction time needs for emergency services. Furthermore, the incorporation of context aware data from violation and parking modules enables the system to change signal behavior in response to surrounding environmental conditions, resulting in a more intelligent and responsive traffic management solution for complicated urban settings.

3.1.1 Computer Vision Pipeline Performance

The computer vision component, based on the YOLO architecture, offers the necessary visual intelligence for efficient vehicle recognition and categorization. Training the model on a localized dataset enables excellent accuracy in identifying vehicle types found on Sri Lankan roadways, such as local buses, taxis, and ambulances. As seen in the Computer Vision Detection Output (Figure 3.1.1), the system has strong localization skills, creating bounding boxes and giving class labels with high confidence scores even in varying lighting circumstances. This visible data acts as additional validation layer for the infrared sensors, resulting in a redundant and extremely accurate traffic monitoring system.

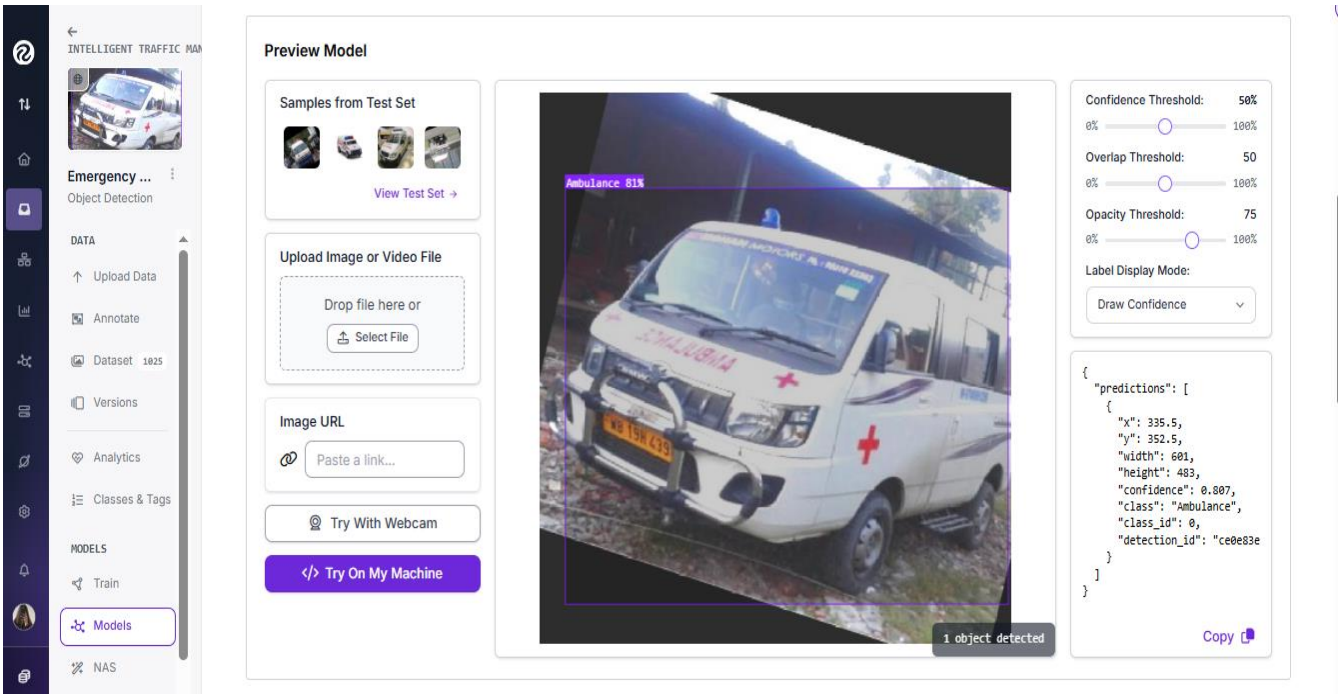

Figure 3.1.1 Computer Vision Detection Output

Analyzing the training metrics reveals a well optimized a learning process for the object identification model, which is a necessary component of the ITMS's visual intelligence layer. The Roboflow Model Training Performance - mAP over Training Epochs plot (Figure 3.1.2) shows a continuous and constant improvement in mean Average Precision as the training epochs progress. This positive trajectory finally achieves a stable plateau, suggesting that the model has correctly detected and learned the differentiating characteristics of the target vehicle classes. The model's good performance on validation data demonstrates that it produces accurate generalizations without over fitting the unique training examples provided.

The continuous decreasing trend in the box loss and class loss curves supports the training process's stability. A decrease in box loss indicates that a model is getting better at finding vehicles and drawing accurate bounding boxes around them. This convergence of high precision and low loss demonstrates that the final AI model is strong enough to perform consistently in real-world detection tasks under a variety of lighting and road situations.

Figure 3.1.2 Roboflow Model Training Performance - mAP over Training Epochs

Statistical validation of the Roboflow pipeline demonstrates its preparedness for implementation within the integrated system architecture. The model earns an exceptional mAP@50 score of 94.7%, demonstrating its superior ability to properly recognize objects at a traditional threshold. In a practical traffic setting, this high score assures that the controller receives high-quality visual data. Accuracy measures at 89.7%, with a significant recall rate of 94.4%. This strong recall is especially important since it ensures that very few vehicles go unnoticed during real operation, preventing the logic from underestimating the true lane density.

The acquired F1 score of 92.0% reflects a solid and essential balance of precision and recall, establishing the vision pipeline as a reliable source of truth. By maintaining this balance, the system avoids both false detections and missing vehicles, ensuring that the decision-making input remains consistent. As a result, the FLC may confidently use these numbers to establish lane priorities and signal timings. This statistical performance shows the computer vision component as a key strength of the ITMS, delivering the dependable data flow required to effectively reduce urban congestion and increase overall junction safety.

3.2 Research Findings

The results of the research show that the ITMS efficiently tackles the limitations of traditional fixed-timer traffic signals by using live data for making decisions. The system's fusion of FL and

ML displays its capacity to process complicated traffic data and offer appropriate signal timing. The FLC implementation enables the system to translate numerical sensor data into linguistic phrases like low or high density, allowing for more human-like reasoning while directing junction flow. This strategy ensures that lane priorities are assigned based on actual demand rather than predetermined schedules, greatly boosting overall vehicle throughput at the intersection.

A important conclusion of this work includes the successful implementation of the priority scoring system in a simulated environment. By awarding particular scores ranging from 10 to 100 based on vehicle numbers, the MCU accurately determines which lane requires the green light first. This dynamic selection mechanism guarantees that lanes with high density are given longer durations, such as 12 seconds, while empty lanes are skipped to save time. The hardware prototype properly displays these changes via LED signal transitions, demonstrating that the logic can be physically implemented to efficiently control real-world traffic situations.

The results add to the importance of the EVP procedure in improving urban road safety. The use of RFID technology enables the system to detect emergency vehicles with high precision, resulting in a quick green light override. This research implies that integrating specialized sensor modules help address emergency service delays in busy locations. Furthermore, the YOLO model's localized training ensures that the AI understand the differences specific vehicle types typical in the Sri Lankan context, providing more accurate input to the traffic logic than generic models do.

Another interesting finding in system monitoring is data communication via IoT services. The system provides a continuous stream of information to the digital dashboard by publishing data to AWS every five seconds using the MQTT protocol. This connectivity provides a clear perspective of junction performance, including live vehicle counts and current signal conditions. The study discovers that low-latency communication are essential for law enforcement to execute remote monitoring and manual overrides when needed, resulting in a comprehensive tool for smarter city management.

Finally, the adaptive nature of the system demonstrates that machine learning can continuously improve the performance of FL rules. The incorporation of learning modules enables the ITMS to evaluate previous patterns and improve its intelligence over time. These findings demonstrate that the ITMS is a realistic, low-cost, and high-performance alternative for updating traffic infrastructure in new urban areas.

3.3 Discussion

The performance of the ITMS indicates that a multi-module input strategy outperforms standard sensor-only systems. By combining IR sensors for quick vehicle counting with a YOLO-based computer vision pipeline, the system reaches a level of accuracy that exceeds ordinary environmental constraints. While IR sensors offer speedy data at the hardware level, the AI vision component checks these counts and accurately classifies vehicle kinds. This duplicate data stream ensures that the FLC receives accurate inputs, resulting in better signal modifications. The achieved mAP@50 score of 94.7% demonstrates that the AI model is highly capable of identifying certain vehicle types in the local context, which is critical for appropriate priority assignment.

The change from inflexible, pre-defined signal timers to a flexible logic system based on language factors was a significant step forward in this research. Converting raw numerical data into phrases like "low," "medium," and "high" flow enables traffic controllers to deal with the inherent volatility of urban road conditions. The priority score system works an effective way for measuring lane relevance in real time. By assigning precise weights to vehicle density, the system maximizes green light durations in high-priority lanes while minimizing idle time in empty lanes. This results in a more flexible traffic flow that can respond to unexpected changes in traffic conditions.

The integration of the EVP protocol with RFID technology addresses significant difficulties in urban emergency management. Standard traffic signals frequently delay the progress of ambulances but the ITMS logic gives these vehicles immediate priority via a dedicated interrupt signal. This function runs independently of overall traffic density, prioritizing emergency response times over standard junction optimization. Furthermore, the use of MQTT for real-time cloud synchronization ensures that this essential activity appears and made available to authorities via the IOT dashboard. This amount of openness and control is critical to improving local traffic administration.

The system's scalability and learning capabilities indicate that it is a possible long-term solution for developing smart cities. Using adaptive learning modules, the ITMS may modify its fuzzy rules based on historical performance data, becoming more intelligent as new traffic patterns are detected. The modular design of the technology, which depends on the MCU and low-cost sensors, makes it financially feasible for large-scale implementation. Future improvements could include integrating higher-resolution capture cameras to help refine the FLC recommendations. Overall,

the debate confirms that the proposed framework provides a strong, data-driven alternative to present traffic control infrastructure by effectively combining hardware reliability and advanced AI intelligence.

3.4 Summary of Each Student's contribution

The individual contributions of IT22363848-Tennakoon I.M.S.R are focused on the design and implementation of the ITMS's core intelligence. The key responsibility was to create the FLC to handle real-time traffic data and optimize signal timing. This component is important for ensuring that the junction runs according to actual vehicle demand rather than scheduled times.

- The data collection procedure collects real-time counts from IR sensors, RFID, and cameras to monitor road conditions. This task entails transforming raw numerical sensor data into fuzzy linguistic phrases like low, medium, or high traffic flow.
- Development the fuzzy inference engine requires creating a thorough set of IF-THEN rules to assess lane priorities. These rules enable the system to process fuzzy inputs and recommend signal change based on calculated priority ratings.
- Integration of defuzzification logic translates fuzzy recommendations into exact commands for the MCU to change green light durations. The durations are set to vary between 5s, 8s, and 12s based on the detected vehicle density.
- The EVP protocol uses RFID technology to prioritize emergency cars in Lane 1. This addition ensures that the system can impose a green light state regardless of the current traffic flow when a valid tag appears.
- Context-aware optimization capabilities enable the system to process input data from external modules, including parking and violation detection. This guarantees that the signal logic remains responsive to the larger context of Sri Lankan traffic conditions.
- The physical hardware prototype exhibits the execution of logic rules by controlling LED signals using an Arduino. Furthermore, the student incorporated adaptive learning capabilities to gradually improve the intelligence of the fuzzy rule foundation over time.

4. CONCLUSION

The creation of this ITMS effectively shows that combining FL and ML delivers a better solution for urban traffic congestion than traditional fixed-timer techniques. The hardware prototype, which relies on an MCU with IR sensors and RFID technology, efficiently captures real-time vehicle density data and converts it into fuzzy linguistic phrases like low, medium, or high traffic flow. The FLC evaluates lane priorities using IF-THEN principles, resulting in dynamic green light durations of 5, 8, or 12 seconds according on actual demand rather than predetermined schedules. Testing demonstrates that the priority scoring algorithm accurately assigns longer durations to congested lanes while reducing idle time for empty roads, resulting in smoother traffic movement at simulated three- and four-way junctions.

The integration of the EVP protocol with RFID represents a significant step forward for emergency response scenarios. When an authorized ambulance tag is spotted, the system sends an immediate interrupt, avoiding the typical fuzzy logic cycle and providing an instant green signal. This capability immediately addresses the safety gaps mentioned in the previous literature, as most systems lack dedicated emergency handling capabilities. Furthermore, the YOLO-based computer vision pipeline achieves 94.7% mAP@50, ensuring accurate vehicle categorization and validating IR sensor data. The AWS cloud architecture, which uses the MQTT protocol, ensures low-latency connectivity, publishing data to the Flutter dashboard every five seconds for real-time monitoring by traffic authorities.

The adaptive learning module allows for continuous improvement of the fuzzy rule base based on past traffic patterns, allowing the system to evolve beyond static logic constraints. Non-functional testing ensures that the system is reliable, secure, and scalable for future expansion across numerous crossings. The ITMS provides a low-cost, high-performance alternative to expensive imported traffic systems that are specifically designed for Sri Lankan Road conditions. This study establishes a practical, validated framework for updating urban traffic infrastructure, reducing vehicle wait times, increasing emergency response efficacy, and contributing to safer, smarter city settings by combining AI with embedded hardware.

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